

phase3_hermetic_issue.sh — Let's Encrypt Phase-3 hermetic issuance proof

Revision: 1 **Last modified:** 2026-07-01T10:08:00Z **Authority:**
Inherits constitution/Constitution.md per §11.4.35. §11.4.18
companion for deploy/letsencrypt/phase3_hermetic_issue.sh.

Overview

Proves — repeatably, from a clean slate, with captured physical evidence — that the custom Caddy image obtains a **real** TLS certificate via the ACME **DNS-01** challenge against a **local Pebble** ACME server, fully offline, and that the project's cert-analyzer verifies the issued cert. It is the anti-bluff core of the Let's Encrypt workstream (§11.4.98 re-runnable · §11.4.107 real-evidence · §11.4.108 runtime-signature).

No bluff: Pebble runs with PEBBLE_VA_ALWAYS_VALID=0 (a **genuine** DNS-01 validation), and the PASS is gated on the analyzer verdicts over the leaf Caddy **actually served**.

Prerequisites

- Rootless Podman + podman-compose (§11.4.161).
- The built image localhost/helix_proxy/caddy-challtestsrv:2.8.4 — run deploy/letsencrypt/build.sh first.
- openssl, curl. Images ghcr.io/letsencrypt/pebble:2.6.0, .../pebble-challtestsrv:2.6.0, docker.io/coredns/coredns:1.11.1 (pulled on first run).
- tests/letsencrypt/cert_analyzer.sh (sourced for the verdicts).

Usage

```
bash deploy/letsencrypt/phase3_hermetic_issue.sh
# run + tear down
KEEP_UP=1 bash deploy/letsencrypt/phase3_hermetic_issue.sh
# leave stack up (Phase-5)
CADDY_HTTPS_PORT=9443 CADDY_HTTP_PORT=9080 bash ...
# override ports
```

Runs under `nice -n 19 ionice -c 3`. Auto-picks a free host port pair if the defaults are busy (this host's :8443 is owned by an operator service — §11.4.174).

Exit codes

Code	Meaning
0	PASS — a real cert was issued via DNS-01 and every cert-analyzer verdict PASSEd
1	FAIL — a product defect (no cert served, or a verdict failed)
2	OPERATOR-BLOCKED / precondition unmet (image not built, podman-compose absent)

What it verifies (the anti-bluff gate)

Over the leaf Caddy served on :CADDY_HTTPS_PORT (SNI proxy.hermetic.test):

- `cert_not_expired` — the served cert is time-valid.
- `cert_san_matches proxy.hermetic.test` — covers the expected name...
- ...and a **negative** check that it does NOT match `evil.example.invalid` (the analyzer is discriminating, not a rubber-stamp).
- `cert_chain_roots_in` — the leaf cryptographically chains to **this run's** Pebble CA, fetched fresh from Pebble's management API (`/roots/0 + /intermediates/0`) — the strongest proof the cert was genuinely issued by this Pebble instance.

Internal behaviour

1. `podman-compose down -v` (fresh Caddy /data — avoids certmagic bug #354).
2. Materialize `pebble-ca/pebble.minica.pem` (Pebble's public trust anchor) via `podman cp`.
3. Boot `pebble + challtestsrv`; resolve `challtestsrv`'s live pod IP.

4. Write `coredns/Corefile` (authoritative `hermetic.test` SOA in the ANSWER section + fall-through forward of the dynamic `_acme-challenge` TXT to that IP); boot CoreDNS.
5. Wait for the Pebble ACME directory; boot caddy — issuance triggers on start.
6. Poll `:CADDY_HTTPS_PORT` for the served leaf with the expected SAN.
7. Fetch this run's Pebble CA; run the cert-analyzer verdicts; write evidence; PASS/FAIL.

Edge cases

- **Image not built / podman-compose absent** → exit 2 (OPERATOR-BLOCKED), never a fake pass.
- **Port :8443 busy** (operator lava-api on this host) → auto-picks 9443/9080.
- **Broken resolver** (e.g. `ACME_RESOLVERS=challtestsrv:8053`, bypassing CoreDNS) → the SOA walk hits NOTIMP, no cert is served, exit 1 — this is exactly what the tests/letsencrypt/phase3_issuance_guard.sh RED_MODE reproduces.
- Cleanup (down -v) runs on every exit path unless `KEEP_UP=1` (§11.4.14).

Internal behaviour — why CoreDNS

certmagic's DNS-01 flow does an SOA walk to determine the zone **before** presenting the TXT; `challtestsrv` answers NOTIMP to SOA. CoreDNS is inserted as an authoritative SOA front (template IN SOA in the ANSWER section) so certmagic accepts `hermetic.test` as the zone, while the dynamic TXT still lives in `challtestsrv` (via forward). Full root-cause: docs/research/letsencrypt_hermetic_20260701/ + docs/research/certmagic_chain_panic_20260701/.

Related scripts

- `deploy/letsencrypt/build.sh` — builds the DNS-01-capable Caddy image (run first).
- `tests/letsencrypt/cert_analyzer.sh` — the verdict functions this script sources.
- `tests/letsencrypt/phase3_issuance_guard.sh` — the §11.4.135 standing regression guard.
- `challenges/scripts/le_phase3_issuance_challenge.sh` — the HelixQA-style Challenge.

Last verified

2026-07-01 — two clean runs (20260701T100119Z, 20260701T100434Z) each issued a real cert (distinct per-run Pebble intermediate CAs 4ce89d / 351928) with all four cert-analyzer verdicts PASS. Evidence: [qa-results/letsencrypt/phase3_issuance/](#).